The Spec Is the New Schematic

Reproducible Station Software with Spec-Driven Development (SDD)

John Carter, N6YU — RSGB Convention 2026

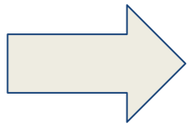
Content and handouts from a presentation at the 2026 RSGB Convention, shared with permission

Presentation Available on GitHub (jcarter-labs)

In case you want to see the presentation and follow along…

Talk repo

Optional Viewing
(now or later)

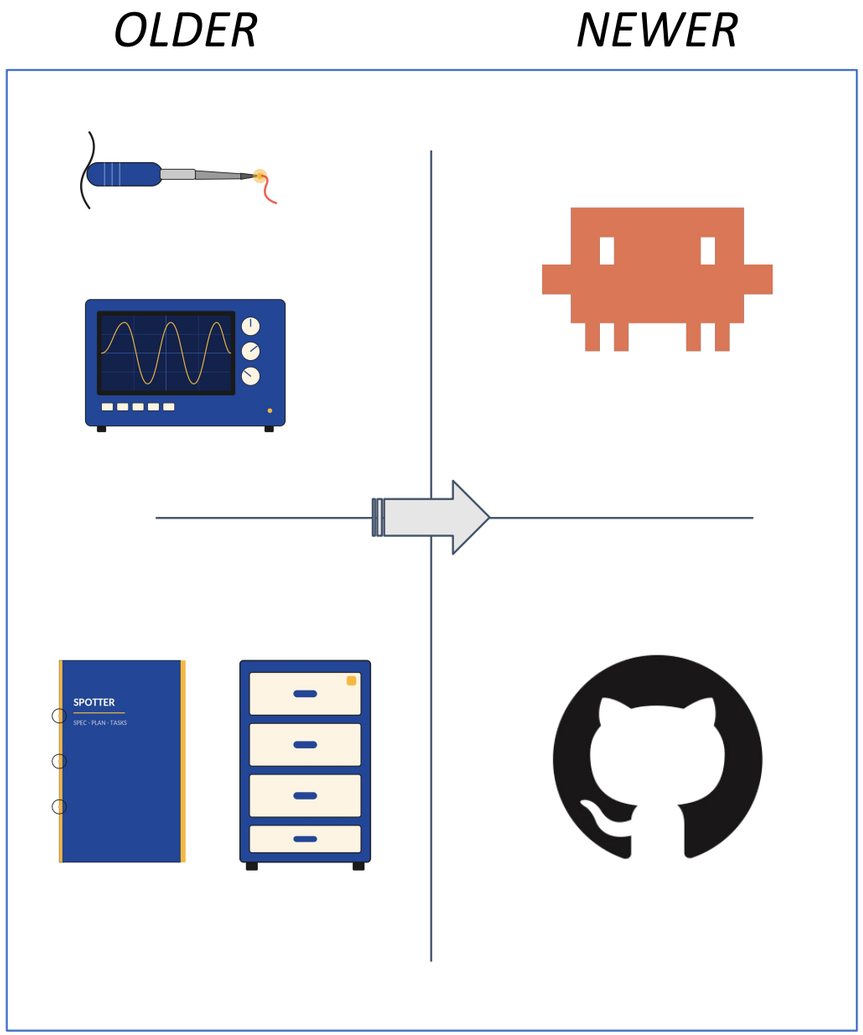




github.com/jcarter-labs/rsgb-sdd-n6yu

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Software Is Replacing Solder



- Today's shack: seems like as much software as hardware
- Adding a feature: **hard** in open source, **impossible** in closed source
- Claude Code is the new way to code
- GitHub is the new file cabinet

Using AI for coding has big drawbacks

- **NOT** Repeatable
- **NOT** Structured

Fortunately good enough for radio projects

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Remote Operation in Mojave Desert

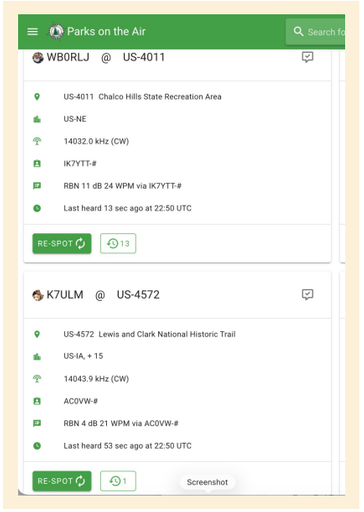


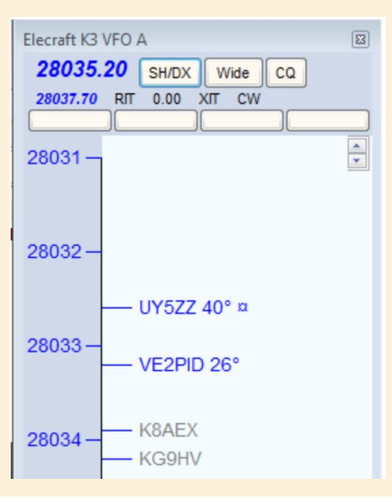


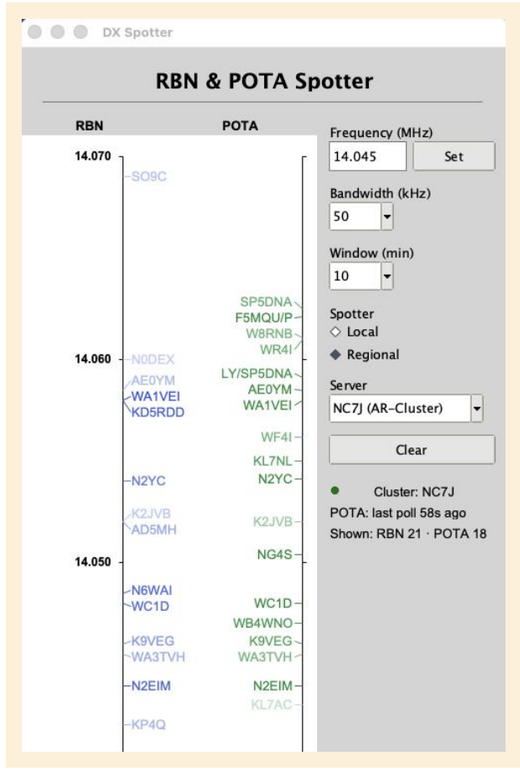
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Why Did I Bother? A Poor Man's Panadapter

- POTA.APP shows maps, and spots for Hunting
  - Interface prioritizes mapping, not bandscope
- N1MM Windows app shows a bandscope, not POTA spots







POTA.APP **N1MM** **SDD DX Spotter - Integrated**

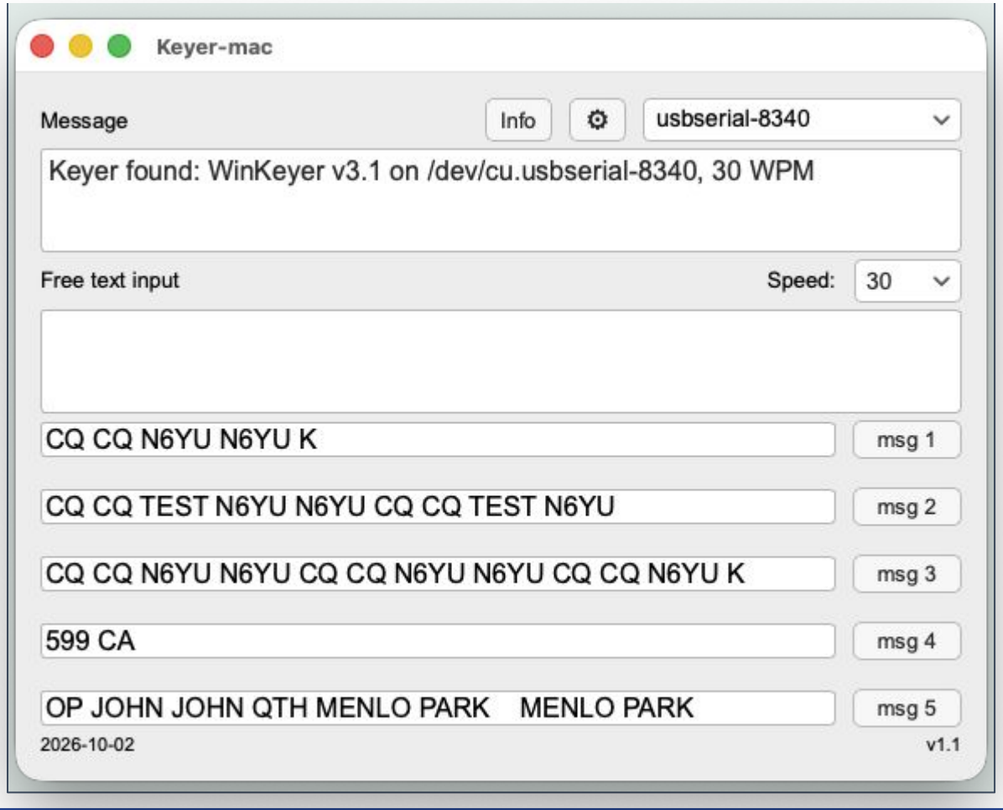
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Possible SDD Projects for Radio

- Platform changes (Win<>Mac<>Linux<>Mobile)
- Open source / GitHub building assistance
- Create code from magazines
- Arduino/Raspberry Pi Projects
- Instrumentation & Automation
  - Data Logging, Temp, Voltage, Current, SWR
- Contact Logging, Tracking Awards, Spotting, Alerts, Band Activity, Antenna Testing
- Station Control: Amps, Switching, Tuners
- Web applications

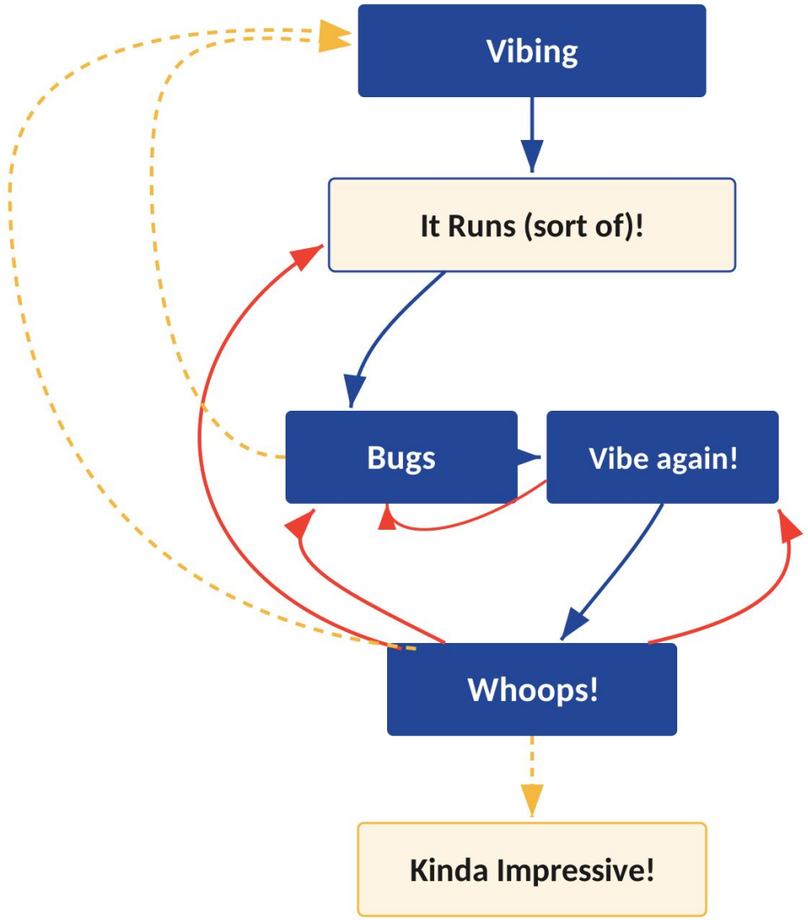
SDD Example of Keyer built from K6GTE

Platform Change: Linux -> Mac



Vibe Coding, Can Be Amazing! POLL

- Vague prompt: “create a spotting app for RBN & POTA” — **silent assumptions**
- Functionality: Display, clusters, UI, parameters — **all guessed, often wrong**
- Testing: Feature tests or just smoke test? — **results in brittle code**
- Now what if you want to come back 6 months from now to tweak it? — **where was that chat?**
- Vibe coding is amazing for what it can accomplish in the first 20% of time — **frequently frustrating**



Key Terms

Agent — AI that edits files, runs code

Claude Code — AI agent in your terminal

CLAUDE.md — Notes Claude reads at startup

Commit — A saved file snapshot with a note

Context window — The AI's working memory

GitHub — Storage website; where files are tracked

Markdown (.md) — Formatted plain text

Masterplan (MP) — **the 4-part Spec for your app**

Masterplan Generator — **Helper to customize MP**

Push — Upload your files to GitHub

Repo — Project folder plus its history

SDD — Spec-Driven Development (Process)

Session — One continuous AI conversation

Slash command — Shortcuts /usage, /status, etc.

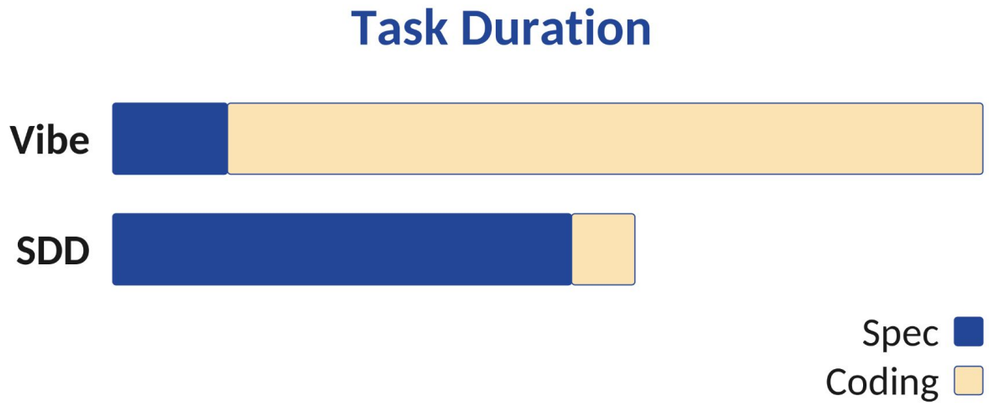
Spec — AI community name for Masterplan

Terminal — Text-only app, no mouse

Token — Word-piece used to meter AI use

A Better Alternative: SDD (Spec-Driven Development)

The model is better at extracting requirements from you than you are at defining them



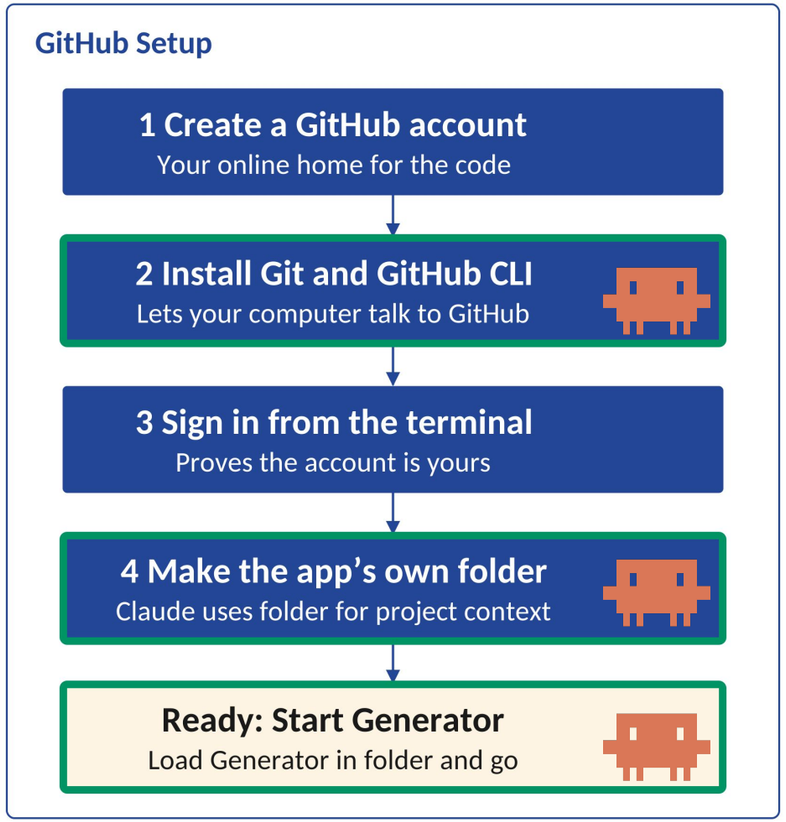
Overall time shorter
More time on the Spec
Less time on Coding & Debug

- ‘Vibing’ wasn’t working for organizations
- SDD Process: Software development process
  - **Create Spec** > Build > Test > Update **Spec**
- Not rigid, many forms, still developing
- The process uses AI to extract requirements

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Preliminaries: Claude Code & GitHub Setup

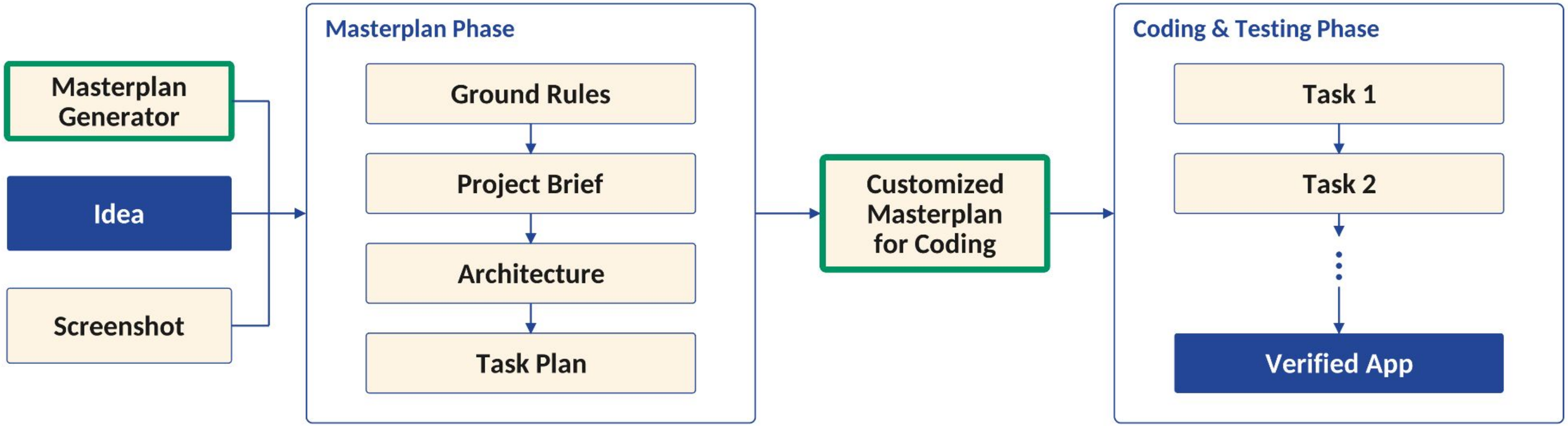
- Why Claude Code (and not Claude, or ChatGPT)
  - Code can loop, span agents, execute programs
  - Chat is a text generator, Code does a lot more
- Install and set up
  - Set up GitHub (cloud), gh (local)
  - Authentication needed (can be hassle)
  - Chrome Connectors (Optional, handy)
- GitHub is a Superpower!
  - Claude can create it, sync it, and create README file for you, organize it, share it…
  - …and then you can find it years later



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SDD in a Nutshell: Masterplan Generator + Idea + Screenshot

- The Masterplan Generator helps you create a Masterplan file tailored to your app
- Cut and paste generator prompts into Claude Code



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Use Generator + Your Idea to Create a Masterplan

Section	Typical Process
1 Ground Rules **how we work**	Use the Generator's Ground Rules prompts as-is: paste into Claude Code, they override all!
2 Project Brief **what it does**	Give Claude your app idea + the Generator will ask you questions about it to customize the Project Brief (UNKNOWN UNKNOWNS)
3 Architecture **built with**	Claude will ask for recommended libraries and approaches, and will DRAFT the Architecture (use example app for input)
4 Task Plan **in what order**	Generator will help structure project into ~5 chunks: Setup, Connections, Core Logic, Main Features, UI (all from the Generator)

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Example: Idea + Screenshot -> Masterplan Generator

My app is a graphical bandmap like N1MM that uses local spotters from an RBN AR6 cluster to display locally received DE spots from anywhere DX on a linear frequency scale, indicating the spot's call sign with a leader line to the frequency, that relies on the deduplication from the cluster.

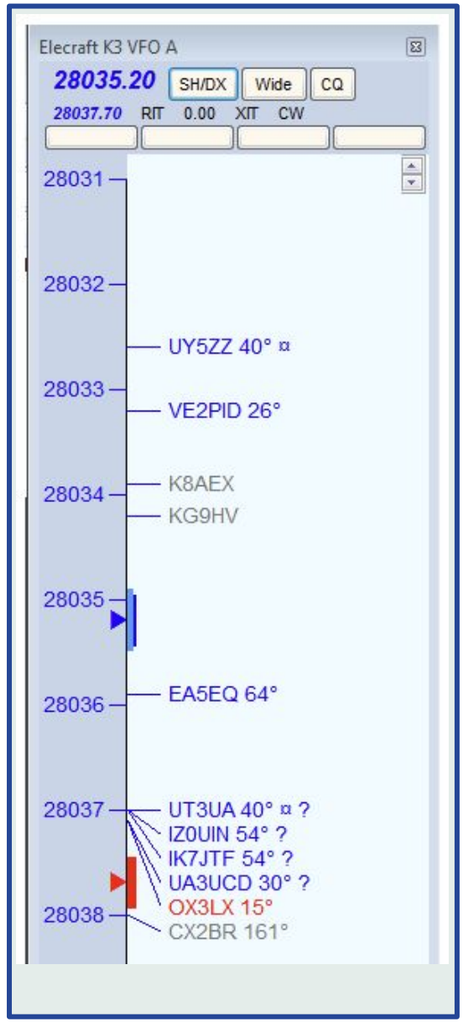
It also displays POTA.app spots on the right frequency scale that are derived via an API from POTA.app.

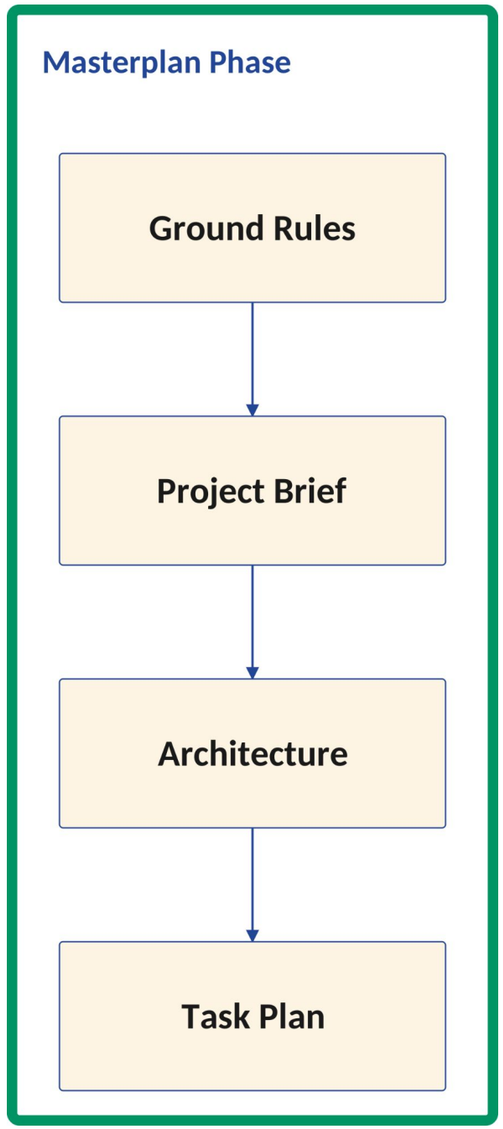
It should have controls like the screenshot, and the window should be resizable. There should be an adjustable fading time so that spots captured 5 - 15 min ago should be faded over time.

You should be able to click a call sign and that copies it into the copy-paste buffer so it can be copied into a logger or QRZ.com for callsign lookup.

It should be built on a Mac and run on a Mac. I would set the center frequency in MHz and bandwidth in kHz, with a default of 50 kHz.

The cluster that you should use is NC7J, and the local spotters are W6YX, AK6RI-1, N6TV, and the regional spotters are K6FOD, WA7LNW, ND7K, K7CO, NG7M, N7VVX, N7TUG, KD7EFG, KW7MM, KW7MM-2. This should be a toggle or radio button as per screenshot.





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The Masterplan Generator

1 Ground Rules *how we work*

PASTE ALL FIVE AT ONCE

RULE Build from masterplan.md, idea.md and my screenshot; borrow language, tools, specs, or open-source code from examples as I choose.

RULE At the start of the build, check tools, libraries, GitHub login, that this folder is its own repo root, and that each data source's host and port can be reached, on my platforms; show pass/fail.

RULE After each change, measure the app against the Project Brief's screen list; show pass/fail.

RULE After each working step: run all tests, show me proof, commit, and push to GitHub. Keep going within a stage; stop only at stage end, on a failed test, after two failed fixes, or for my decision.

RULE When code and masterplan disagree, propose only major changes, one line each; update the masterplan after I approve.

2 Project Brief *what the app does*

ONE AT A TIME

DRAFT Summarize what my app does in a few sentences, from idea.md, my screenshot, and any example app.

DRAFT From my screenshot, list screen elements and controls, with where each sits.

DRAFT For each feature in idea.md, describe what the user does and sees, with testable ranges where possible.

DRAFT List where my app's data comes from, and how we'll check each source works, including after a reconnect. Send every server setting on each connect; never rely on settings from earlier sessions.

USER INPUT Ask which features from idea.md my app must do this iteration, and which wait; suggest if I'm unsure.

3 Architecture *what it's built with*

ONE AT A TIME

USER INPUT Recommend a language and tools, Python unless my example suggests better; explain each and let me choose.

DRAFT List what must be installed on each platform in idea.md, and how to check each works.

RULE Keep the app responsive while it works, never stuck waiting on data or input.

DRAFT Propose a handful of well-structured modules, one line each, each testable on its own.

RULE Keep a short list of known limitations in the masterplan; update it as we go.

4 Task Plan *in what order*

ONE AT A TIME

DRAFT Break the build into about 5 stages: environment (including measuring my screenshot with a script), data connections tested on live servers and ending with a bare window showing live data, core logic, features, and UI. Give each stage sub-steps and one done-when line.

DRAFT For each sub-step, say how we'll test it and what result means it works.

RULE Test connections to outside data with real servers before building screens that depend on them.

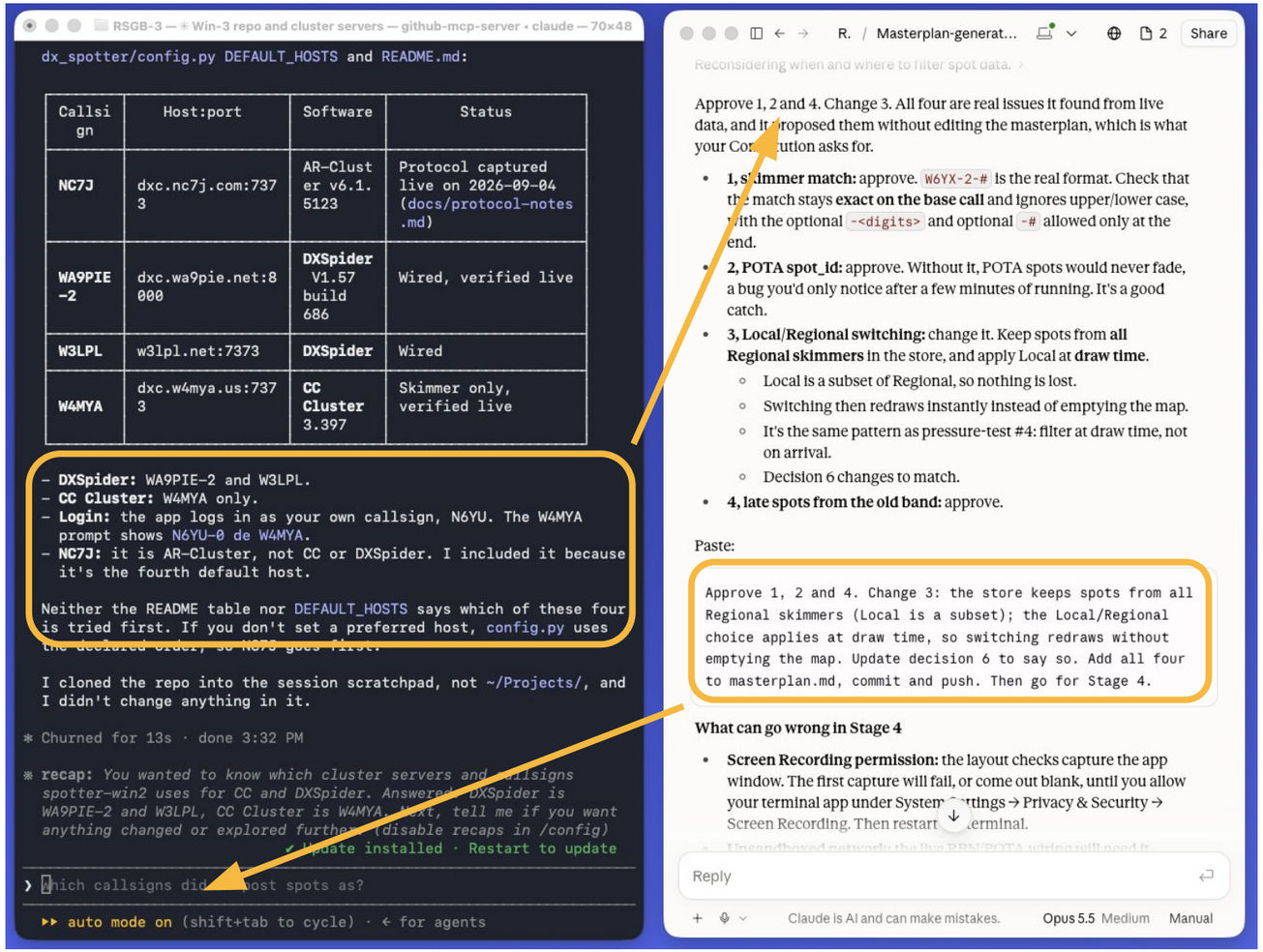
RULE Get a simple version running early, then add features one at a time, testing each.

DRAFT Add a final step: review what went wrong and propose masterplan updates for my approval.

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Claude Code with Claude Desktop as Helper (Screenshot)

1. Generator causes Code to generate a question I don't understand



Claude Code **Claude Desktop**

2. I paste the question in Desktop.

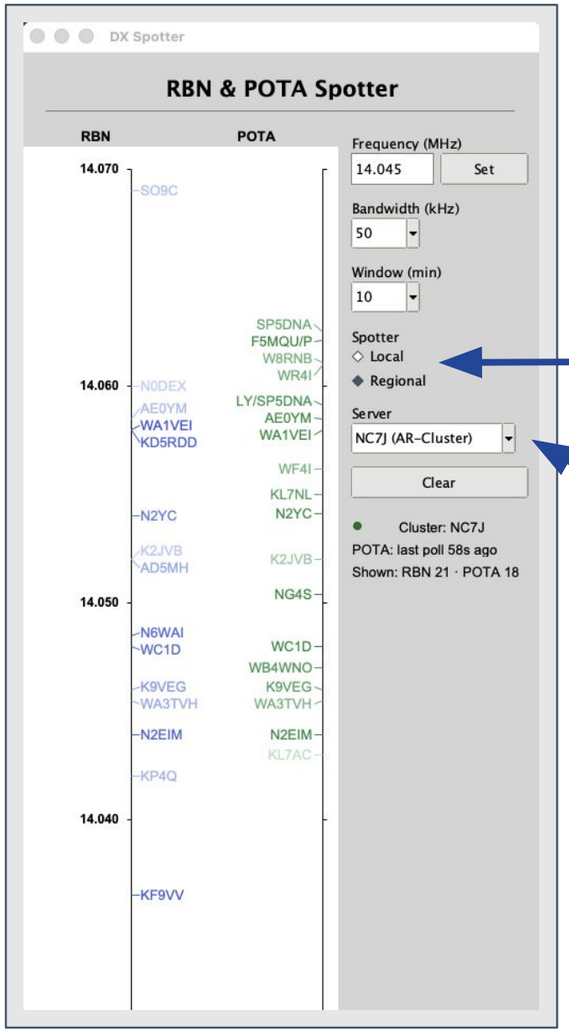
3. I guess an answer, but ask Claude Desktop for help

4. Paste answer back into Code

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It Works! Multiplatform Screenshots

Mac



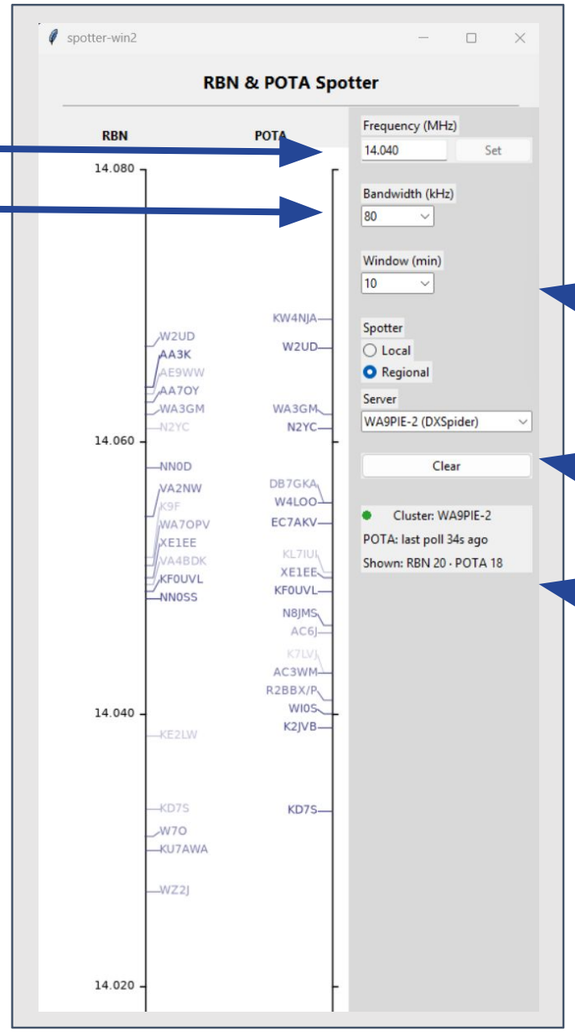
Center Frequency

Bandwidth

Regional or Local Spotters

AR6 CC Cluster DX Spider

Win



Window (min)

POTA Status

Spot Counts

Linux (not shown, looks exactly the same)

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Conclusion - It Works, Give it a Try!

1. Get an agentic AI app (Claude Code or equivalent)
2. Get a GitHub account & use it for one project
3. Try out SDD with Masterplan Generator

Talk repo

Slides, Masterplan Generator

github.com/jcarter-labs/rsgb-sdd-n6yu

Time for Your Road Trip! Your Idea and…
…Transportation Provided by Claude Code and GitHub



Questions?

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Appendix - References

- "Spec-Driven Development: From Code to Contract in the Age of AI Coding Assistants", Deepak Babu Piskala: https://arxiv.org/pdf/2602.00180
- Claude Code Tutorials (Anthropic short courses): https://anthropic.skilljar.com/
- Field Spotter (Ian Renton): https://git.ianrenton.com/ian/field-spotter
- DeepLearning.AI (Practical short course on application of SDD): Spec-Driven Development with Coding Agents - DeepLearning.AI
- Slides, generator, references: github.com/jcarter-labs/rsgb-sdd-n6yu

Find out more…

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